10. A transportation network contains cities connected by roads with different costs. Write a C program implementing Dijkstra's Shortest Path Algorithm that accepts the number of vertices, weighted adjacency matrix and source vertex, computes the minimum distance from the source to every other vertex, and displays each destination with its shortest distance. Test it using at least five vertices.

Source code :-

```
#include <stdio.h>
#define INF 9999
int main()
{
    int cost[20][20], dist[20], visited[20] = {0};
    int n, src, i, j, count, min, u;
    printf("Enter number of vertices: ");
    scanf("%d", &n);
    printf("Enter weighted adjacency matrix (0 = no road):\n");
    for (i = 0; i < n; i++)
    {
        for (j = 0; j < n; j++)
        {
            scanf("%d", &cost[i][j]);
            if (cost[i][j] == 0 && i != j)
                cost[i][j] = INF;
        }
    }
    printf("Enter source vertex (0 to %d): ", n - 1);
    scanf("%d", &src);
    if (src < 0 || src >= n)
    {
        printf("Invalid source vertex\n");
        return 0;
    }
    for (i = 0; i < n; i++)
        dist[i] = INF;
    dist[src] = 0;
    for (count = 0; count < n - 1; count++)
    {
        min = INF;
        u = -1;
        for (i = 0; i < n; i++)
        {
            if (visited[i] == 0 && dist[i] < min)
            {
                min = dist[i];
                u = i;
            }
        }
        if (u == -1)
            break;
        visited[u] = 1;
        for (j = 0; j < n; j++)
        {
            if (visited[j] == 0 && cost[u][j] != INF &&
                dist[u] + cost[u][j] < dist[j])
                dist[j] = dist[u] + cost[u][j];
        }
    }
    printf("\nVertex\tShortest Distance from %d\n", src);
    for (i = 0; i < n; i++)
    {
```

```
        if (dist[i] == INF)
            printf("%d\t\tNot reachable\n", i);
        else
            printf("%d\t\t%d\n", i, dist[i]);
    }
    return 0;
}
```

Output :-

Test 1: 5 vertices

```
Enter number of vertices: 5
Enter weighted adjacency matrix (0 = no road):
0 10 0 5 0
0 0 1 2 0
0 0 0 0 4
0 3 9 0 2
7 0 6 0 0
Enter source vertex (0 to 4): 0

Vertex  Shortest Distance from 0
0               0
1               8
2               9
3               5
4               7
```

Test 2: 6 vertices

```
Enter number of vertices: 6
Enter weighted adjacency matrix (0 = no road):
0 7 9 0 0 14
7 0 10 15 0 0
9 10 0 11 0 2
0 15 11 0 6 0
0 0 0 6 0 9
14 0 2 0 9 0
Enter source vertex (0 to 5): 0

Vertex  Shortest Distance from 0
0               0
1               7
2               9
3               20
4               20
5               11
```